

# Rings: An Algebraic Structured Peer-to-Peer Network for Decentralized Identity and Resource Ownership

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## Abstract

Rings is an implemented algebraic structured peer-to-peer network for decentralized identity, resource ownership, transport, storage, and application protocols. The project starts from a concrete failure mode in contemporary Internet systems: application servers, DNS names, relay services, and platform accounts often become the authority that identities and data must trust. Rings instead treats identities and resources as algebraic objects over a structured CorrectChord overlay, with browser and native runtimes connected through WebRTC/WebAssembly adapters and cryptographic protocols kept on carriers separate from the routing identifier space. The core object is

$$\mathcal{R} \setminus \setminus f = (R, \mathcal{R}_N, \mathbf{Set}/R, \Omega_N, \mathsf{T}_A^\mathcal{E}, \Pi, \mathbb{G}, \mathcal{X})$$

with

$$\begin{aligned} R &= \mathbb{Z}/2^{160}\mathbb{Z}, \\ \mathcal{R}_N &= \text{Correct-Chord overlay object}, \\ \Omega_N(X, H) &= (X, \text{owner}_N \circ H), \\ \tau_\nu &: S_\nu \times I_\nu \rightarrow \mathsf{T}_{A_\nu}^{\mathcal{E}_\nu}(S_\nu \times O_\nu). \end{aligned}$$

$$\{D, V, M, H, Q, W\} \subset \text{Obj}(\mathbf{Set}/R), \quad \text{CryptoCarrier} \cap R = \emptyset.$$

## I. Introduction

Centralized application platforms simplify deployment, but they also concentrate identity, naming, storage, and traffic control at service operators. Cloud and platform centralization therefore becomes a privacy, availability, and autonomy risk for users who need direct communication and durable control over their data [1]–[3]. The Rings project is motivated by this observation: a sovereign network should let a participant own an identity, establish a session, route to another identity or resource, and run application protocols without turning a single application server into the root of authority.

Rings is implemented as a network, but this paper is organized around the algebraic structure that makes the implementation coherent. It states the ideal carrier, ownership, transition, and safety laws that the implementation realizes and continues to refine. The older intuition was that Decentralized Identifiers (DIDs) directly form a finite ring suitable for all routing and cryptographic operations. The refined model is more precise: Chord identifiers form an additive circular carrier for placement, interval order, replica selection, and finger targets; threshold schemes, signatures, encryption, and zero-knowledge protocols use their own fields or groups.

The resulting design has three practical consequences. First, DIDs and virtual DIDs (VIDs) share the same 160-bit placement surface, so identities, mailboxes, service

descriptors, chunks, subrings, and application rendezvous points can be routed by one owner function. Second, Rings builds on Chord and CorrectChord rather than replacing the overlay literature: successor lists, predecessor relations, join, rectify, and stabilize are the maintenance substrate on which Rings semantics are layered [4], [5]. Third, state transitions return typed effects instead of directly performing network IO, which allows the overlay to be reasoned about as a pure state machine and then interpreted by browser or native runtimes.

## II. Problem Statement and Design Goals

Rings addresses a narrow but recurring gap between existing decentralized systems. Kademlia-derived systems provide scalable lookup, but the XOR metric does not by itself give Rings the circular ownership law needed for successor lists, replicated ownership, and interval-based placement. Relay networks and mixnets improve metadata resistance, but they are not structured P2P ownership systems. Blockchains provide global ordering, but embedding consensus in every message or storage operation would turn Rings into a ledger rather than an algebraic structured peer-to-peer substrate.

The problem is therefore to specify a network object

$$\mathcal{R} \setminus \setminus f$$

that simultaneously supports:

- 1) self-certifying identity and delegated sessions;
- 2) structured lookup and resource ownership;
- 3) browser-native and native transport surfaces;

- 4) application protocols with typed side effects;
- 5) cryptographic protocols over correct carriers;
- 6) bounded algebraic and operational obligations for routing and state repair.

This paper makes the following contributions. It presents Rings as an algebraic structured peer-to-peer network, not merely as an overlay lookup routine. It models the Rings identifier space as the additive cyclic group  $R = \mathbb{Z}/2^{160}\mathbb{Z}$ , not as a general finite field. It lifts resource placement into **Set**/ $R$ , so each protocol object carries an explicit map into the overlay. It uses CorrectChord as the maintenance law before introducing Rings-specific replication, storage, mailbox, relay, and ranking semantics. It gives a writer effect monad for DHT transitions and a TLA-style state relation for model checking and trace replay. These algebraic and operational artifacts are the core of the paper; the surrounding text explains why those artifacts exist and how they relate to prior systems.

### III. Algebraic System Model

Assumption 1 (System scope).

$$\begin{aligned}
\text{fault} &= \text{fail-stop}, \\
\text{auth}(m, \sigma, k) &= V(k, \sigma, m), \\
\text{net} &= \text{async}, \\
\text{io} &= I_X : A_X^* \rightarrow \text{IO}, \\
\text{consensus} &\notin \text{Overlay}, \\
\text{order} &= \text{SidecarWitness}, \\
\text{crypto\_ops} \cap \text{route\_ops} &= \emptyset.
\end{aligned}$$

Definition 1 (Identifier carrier).

$$\begin{aligned}
M &= 2^{160}, \\
R &= \mathbb{Z}/M\mathbb{Z}, \\
\delta_a(x) &= (x - a) \bmod M, \\
(a, b]_R &= \{x \in R : 0 < \delta_a(x) \leq \delta_a(b)\}, \\
\text{RouteOps}_R &= \{0, +, -, \delta_a, (\_, \_]_R\}, \\
\text{Mult}_R &\notin \text{RouteOps}_R.
\end{aligned}$$

Definition 2 (Live topology).

$$\begin{aligned}
N &\subset R, \quad N \neq \emptyset, \\
\text{owner}_N(k) &= \arg \min_{u \in N} \delta_k(u), \\
\text{pred}_N(n) &= \arg \min_{p \in N \setminus \{n\}} \delta_p(n), \\
\text{Succ}_N^q(n) &= \text{take}_q(\text{sort}_{\delta_n}(N \setminus \{n\})), \\
\delta_a|_N &= \text{injective}.
\end{aligned}$$

Assumption 2 (Correct-Chord stable base).

$$\begin{aligned}
r &= |\text{Succ}_N^r(n)|, \\
\text{Steady}(N) &\Rightarrow |N| \geq r + 1, \\
|N| < r + 1 &\Rightarrow \text{Init}(N), \\
\text{Maintain} &= \{\text{join}, \text{rectify}\} \\
&\quad \cup \{\text{stabilize}\}_{\text{CorrectChord}}.
\end{aligned}$$

This is the CorrectChord stable base of Zave [5].

Definition 3 (Replica set).

$$\text{Rep}_N^r(k) = \{\text{owner}_N(k)\} \cup \text{set}(\text{Succ}_N^{r-1}(\text{owner}_N(k))).$$

Definition 4 (Rings overlay object).

$$\begin{aligned}
\mathcal{R}_N &= (R, N, \text{owner}_N, \text{pred}_N, \text{Succ}_N^r, \text{Rep}_N^r, F, E), \\
F &\subseteq N \times \mathbb{N} \times N, \\
E &= \prod_{v \in \text{VID}} E_v, \\
\text{SafetyRoot}(\mathcal{R}_N) &= (\text{pred}_N, \text{Succ}_N^r), \\
\text{PerfWitness}(\mathcal{R}_N) &= F.
\end{aligned}$$

Proposition 1 (Rotation equivariance).

$$\begin{aligned}
t \in R, \quad N + t = \{n + t : n \in N\} &\Rightarrow \\
\text{owner}_{N+t}(k + t) &= \text{owner}_N(k) + t.
\end{aligned}$$

Proof.

$$\begin{aligned}
\forall n \in N. \quad \delta_{k+t}(n + t) &= ((n + t) - (k + t)) \bmod M \\
&= \delta_k(n).
\end{aligned}$$

□

Constraint 1 (Carrier separation).

$$\begin{aligned}
\text{PlacementOps} &= \{0, +, -, \delta_a, (a, b]_R\} \\
&\quad \cup \{\text{owner}_N, \text{Rep}_N^r\}, \\
\text{CryptoOps} &= \{\times, ^{-1}, \cdot_{\text{scalar}}, \text{interp}\} \\
&\quad \cup \{\text{pair}, \text{subgroup}\}, \\
\text{PlacementOps} \cap \text{CryptoOps} &= \emptyset, \\
\text{CryptoCarrier} &\in \{\mathbb{F}_q, \mathbb{G}, \mathbb{Z}_q\}.
\end{aligned}$$

| Object         | Carrier or law                               |
|----------------|----------------------------------------------|
| Chord IDs      | additive cyclic group $R$                    |
| DID proof      | signature verifier and transcript law        |
| VID            | $H_\nu : X_\nu \rightarrow R$ then owner map |
| placement      | join semilattice or explicit finalizer       |
| Storage merge  | finite field or curve group                  |
| E2E encryption |                                              |
| Sequencing     | partial order over witness domains           |
| Effects        | writer monad over action lists               |

Table I

Carrier boundaries in the Rings model

### IV. Architecture As Interfaces

Definition 5 (Layer tuple).

$$\mathcal{A} = (I, T, O, X, P, L)$$

$$\begin{array}{l|l}
I & (DID, \Pi, \text{Session}) \\
T & (C, A_T, \text{offer}, \text{answer}, \text{send}, \text{recv}) \\
O & \mathcal{R}_N \\
X & (S_X, A_X, \text{cap}, I_X) \\
P & \{\mathcal{P}_\nu\}_{\nu \in \text{Namespace}} \\
L & \text{App} \circ P
\end{array}$$

Definition 6 (Identity proof).

$$\Pi = (K, \Sigma, V, D, \text{enc}, \text{ttl})$$

$$\begin{aligned}
V &: K \times \Sigma \times B^* \rightarrow \{\top, \perp\}, \\
D &: K \rightarrow R, \\
\text{enc} &: \Sigma \rightarrow B^*, \\
\text{ttl} &: \Sigma \rightarrow \text{Time}.
\end{aligned}$$

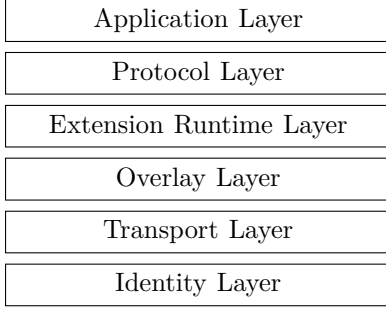


Figure 1. Layers of Rings Network

Invariant 1 (Session authority).

$$d \in X_D, \quad s \in K.$$

$$\text{Session}(d, s, e) \Rightarrow V(k_d, \sigma_{d \rightarrow s}, d \| s \| e) = \top \wedge \text{now} < e.$$

$$\text{HopCheck}(d, s, e) \equiv \text{Session}(d, s, e).$$

Definition 7 (Transport interface).

$$T = (C, \text{offer}, \text{answer}, \text{send}, \text{recv})$$

$$A_T = \{\text{Offer}, \text{Answer}, \text{Ice}, \text{Open}, \text{Send}, \text{Recv}, \text{Close}\}.$$

$$I_T : A_T^* \rightarrow \text{IO}_{WebRTC} \cup \text{IO}_{Native}.$$

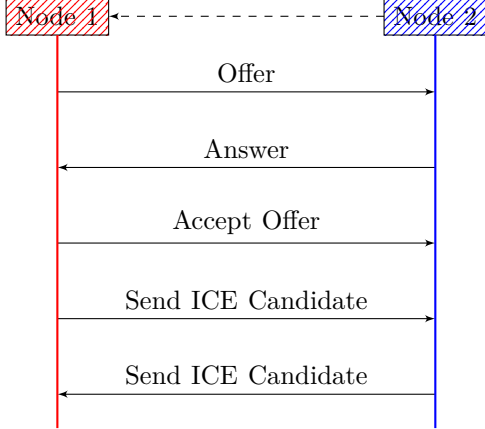


Figure 2. Standard SDP/ICE exchange

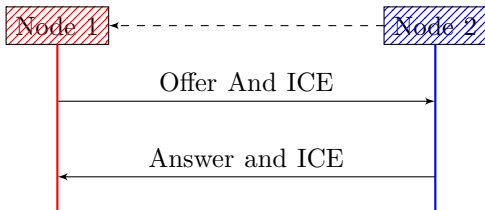


Figure 3. Rings single round-trip SDP/ICE exchange

Definition 8 (Runtime interpreter).

$$\mathcal{X} = (S_X, A_X, \text{cap}, I_X)$$

$$\begin{aligned} \text{cap} &\subseteq A_X, \\ I_X &: A_X^* \rightarrow \text{IO}, \\ I_X(\epsilon) &= \text{Noop}, \\ I_X(\alpha \cdot \beta) &= I_X(\alpha); I_X(\beta). \end{aligned}$$

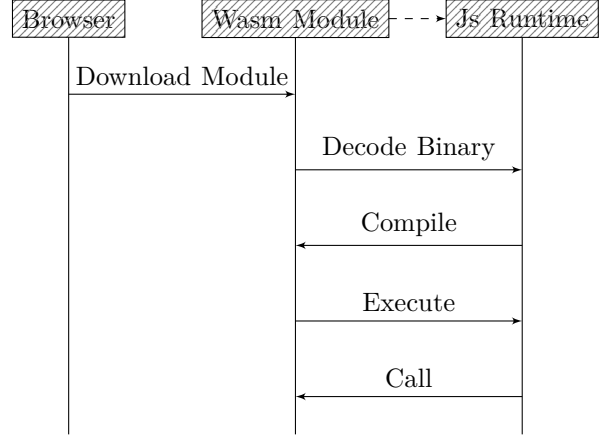


Figure 4. Wasm module execution boundary

Obligation 1 (Runtime confinement).

$$\begin{aligned} \tau &: S \times I \rightarrow \mathcal{E} + (S \times O \times A_X^*), \\ \text{emit}(\tau(s, i)) &\subseteq \text{cap}, \\ a \notin \text{cap} &\Rightarrow I_X(a) = \text{Reject}(a). \end{aligned}$$

## V. Category and Effects

Definition 9 (Namespace object).

$$\begin{aligned} \nu &\in \{D, V, M, H, U, Q, W\}, \\ H_\nu &: X_\nu \rightarrow R, \\ (X_\nu, H_\nu) &\in \mathbf{Set}/R. \end{aligned}$$

Definition 10 (Placement functor).

$$\begin{aligned} \Omega_N &: \mathbf{Set}/R \rightarrow \mathbf{Set}/N, \\ (X, H) &\mapsto (X, \text{owner}_N \circ H). \end{aligned}$$

$$P_\nu^N = \text{owner}_N \circ H_\nu : X_\nu \rightarrow N.$$

Proposition 2 (Placement functoriality).

$$\begin{aligned} f : X_\nu \rightarrow X_\mu, \quad H_\mu \circ f &= H_\nu. \\ P_\mu^N \circ f &= P_\nu^N. \end{aligned}$$

Proof.

$$\text{owner}_N \circ H_\mu \circ f = \text{owner}_N \circ H_\nu.$$

□

Definition 11 (Writer effect monad).

$$\begin{aligned} A^* &= \mathbf{FreeMonoid}(A), \\ \mathsf{T}_A(X) &= X \times A^*, \\ \eta(x) &= (x, \epsilon), \\ (x, \alpha) \gg= f &= \text{let } f(x) = (y, \beta) \text{ in } (y, \alpha \cdot \beta). \end{aligned}$$

Proposition 3 (Effect associativity).

$$(h^* \circ g^*) \circ f^* = h^* \circ (g^* \circ f^*).$$

Proof.

$$((\alpha \cdot \beta) \cdot \gamma) = (\alpha \cdot (\beta \cdot \gamma)).$$

□

Definition 12 (Typed failure stack).

$$\begin{aligned} \mathsf{T}_A^\mathcal{E}(X) &= \mathcal{E} + (X \times A^*). \\ \pi_A(e \in \mathcal{E}) &= \epsilon, \\ \pi_A(x, \alpha) &= \alpha, \\ \text{RecoverableReport} &\in A. \end{aligned}$$

Definition 13 (Protocol object).

$$\begin{aligned} \mathcal{P}_\nu &= (X_\nu, S_\nu, I_\nu, O_\nu, A_\nu, \mathcal{E}_\nu, H_\nu, \tau_\nu) \\ \tau_\nu : S_\nu \times I_\nu &\rightarrow \mathsf{T}_{A_\nu}^{\mathcal{E}_\nu}(S_\nu \times O_\nu). \end{aligned}$$

Definition 14 (Protocol morphism).

$$\begin{aligned} \Sigma : \mathcal{P}_\nu &\rightarrow \mathcal{P}_\mu = (\sigma_X, \sigma_S, \sigma_I, \sigma_O, \sigma_\mathcal{E}, \sigma_A^*) \\ \sigma_A^* &: A_\nu^* \rightarrow A_\mu^*, \\ \sigma_A^*(\alpha \cdot \beta) &= \sigma_A^*(\alpha) \cdot \sigma_A^*(\beta), \\ H_\mu \circ \sigma_X &= H_\nu, \\ \mathsf{T}_{\sigma_A^*}^{\sigma_\mathcal{E}}(\sigma_S \times \sigma_O) \circ \tau_\nu &= \tau_\mu \circ (\sigma_S \times \sigma_I). \end{aligned}$$

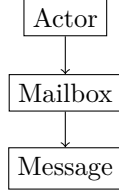


Figure 5. Actor Model

## VI. Chord Overlay

$$\begin{aligned} \text{Base} &= \text{Chord}, \\ \text{Maintenance} &= \text{CorrectChord}, \\ \text{Object} &= \mathcal{R}_N, \\ \text{Witnesses} &= \{\text{Algorithms 1, 2, 3}\}. \end{aligned}$$

This section fixes Chord as the base protocol and CorrectChord as the maintenance law [4], [5].

Definition 15 (Local node state).

$$\begin{aligned} S_n &= (p_n, L_n, F_n, E_n) \\ p_n &\in N \cup \{\perp\}, \\ L_n &\in N^{\leq r}, \\ F_n &: \mathbb{N} \rightarrow N, \\ E_n &\subseteq E, \\ p_n = \text{pred}_N(n), \quad L_n &= \text{Succ}_N^r(n). \end{aligned}$$

Definition 16 (Finger target).

$$\begin{aligned} \text{target}_i(n) &= n + 2^i, \\ \text{finger}_i(n) &= \text{owner}_N(\text{target}_i(n)), \\ F_n(i) &\in \{\perp, \text{finger}_i(n)\}. \end{aligned}$$

Invariant 2 (Ring fixpoint).

$$\begin{aligned} \text{Inv}_{\text{ring}}(N, S) &\equiv \forall n \in N. p_n = \text{pred}_N(n) \\ &\quad \wedge L_n = \text{Succ}_N^r(n). \end{aligned}$$

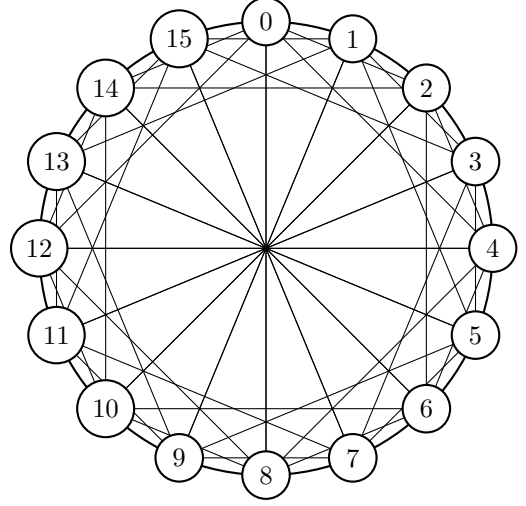


Figure 6. Chord algorithm, lookup protocol

### Algorithm 1 DHT Join Transitions

---

```

1: function BeginJoin( $n, known$ )
2:    $pred[n] \leftarrow none$ 
3:    $a \leftarrow FindSuccessor(n.did)$ 
4:   return  $Remote(known, a)$ 
5: function InstallSuccessor( $n, s$ )
6:    $succ[n] \leftarrow take(r, [s])$ 
7:   return  $Remote(s, QuerySuccList(s))$ 
8: function InstallSuccList( $n, s, list$ )
9:    $succ[n] \leftarrow take(r, s :: list)$ 
10:  if  $head(succ[n]) \neq none$  then
11:     $h \leftarrow head(succ[n])$ 
12:     $notify \leftarrow Notify(h, n.did)$ 
13:     $sync \leftarrow SyncStorage(h)$ 
14:    return  $Multi(notify, sync)$ 
15:  else
16:    return  $Noop$ 

```

---

Invariant 3 (Lookup owner preservation).

$$\begin{aligned} \text{Steady}(N) \wedge \text{Inv}_{\text{ring}}(N, S) \\ \Rightarrow \text{Lookup}(S, k) \Downarrow \text{owner}_N(k). \\ \Downarrow = \text{Reach}_{A_D}(2, 3). \end{aligned}$$

## VII. DID and Resource Algebra

Definition 17 (DID namespace).

$$\mathcal{D} = (X_D, H_D, \Pi, \text{Session}), \quad H_D : X_D \rightarrow R.$$

$$\begin{aligned} D(k) &= H_D(k), \\ \text{owner}_N(D(k)) &\in N, \\ \text{Session} &\subseteq X_D \times K \times \text{Time}. \end{aligned}$$

Definition 18 (VID namespace).

$$\begin{aligned} \mathcal{V} &= (X_V, H_V, \text{Auth}_V), \\ H_V &: X_V \rightarrow R, \\ \text{PrivateKey}(H_V(x)) &= \perp, \\ \text{Auth}_V(op, x, \sigma, caps) &\in \{\top, \perp\}. \end{aligned}$$

---



---

```

1: function Lookup(key, node)
2:   s  $\leftarrow$  head(succ[node])
3:   if s = none then
4:     return Repair(QuerySuccList(node))
5:   if key  $\in$  (node.did, s] then
6:     return Local(s)
7:   next  $\leftarrow$  closest_preceding_node(node, key)
8:   if next = none then
9:     return Repair(QuerySuccList(s))
10:  else
11:    a  $\leftarrow$  FindSuccessor(key)
12:    return Remote(next, a)

```

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**Algorithm 5** ElGamal Encryption and Decryption

---

```

1: Input: Message  $m$ , private key  $x \in \mathbb{Z}_q$ , public key
    $h = g^x$ , generator  $g$ 
2: Output: Encrypted message  $(c_1, c_2)$ 
3: procedure Encryption
4:   Choose a random integer  $k \in \mathbb{Z}_q$ 
5:   Compute  $c_1 = g^k$ 
6:   Compute  $c_2 = m \cdot h^k$ 
7:   return  $(c_1, c_2)$ 
8: procedure Decryption
9:   Compute  $m = c_2 \cdot (c_1)^{-x}$ 
10:  return  $m$ 

```

---

Definition 26 (Secret sharing placement).

$$\begin{aligned}
P &\in \mathbb{F}_q[X], \\
sh_i &= (i, P(i)) \in \mathbb{F}_q^2, \\
H_S(sh_i) &\in R, \\
\text{reconstruct}(\{sh_i\}) &= P(0) \quad (\mathbb{F}_q\text{-interp.})
\end{aligned}$$

This is Shamir reconstruction over the separate field carrier [7].

Definition 27 (Relay frame).

$$\begin{aligned}
r &= (\text{path}, \text{dst}, \nu, \text{session}, \text{payload}, \text{report}). \\
\text{RelayValid}(r) &= \text{SessionValid}(\text{session}) \\
&\quad \wedge \text{dst} \in R, \\
\text{SplitVocabulary} &= \text{MSRP}.
\end{aligned}$$

The split vocabulary follows MSRP [8].

---

**Algorithm 6** End-to-End Chunking Algorithm

---

```

1: procedure E2E Chunking
2:   Input: data, chunk size
3:   Output: chunks
4:   chunks  $\leftarrow []$ 
5:   total-chunks  $\leftarrow \text{ceil}(\text{len}(\text{data})/\text{chunk-size})$ 
6:   for  $i$  in range(total-chunks) do
7:     chunk  $\leftarrow \text{data}[i*\text{chunk-size} : (i+1)*\text{chunk-size}]$ 
8:     append chunk to chunks
9:   return chunks

```

---

Invariant 5 (Chunk reassembly).

$$\begin{aligned}
\text{complete}(m) &\Leftrightarrow \forall i < \text{total}(m). \exists! c_i \\
&\quad \wedge \sum_i |c_i| \leq B_m \\
&\quad \wedge \text{now} < \text{ttl}(m).
\end{aligned}$$

### IX. Ordering, Ranking, and Security

Definition 28 (Sidecar witness).

$$\begin{aligned}
w &= (\text{domain}, \text{root}, \text{height}, \text{time}, \text{finality}). \\
c &= H(\text{msg}), \\
\text{Witness}(\text{msg}, w) &\Leftrightarrow \text{Include}(c, w.\text{root}) = \top \\
&\quad \wedge \text{Final}(w) = \top.
\end{aligned}$$

Definition 29 (Witness order).

$$\begin{aligned}
a &\prec b \Leftrightarrow \text{domain}(w_a) = \text{domain}(w_b) \\
&\quad \wedge \text{time}(w_a) < \text{time}(w_b). \\
\chi &: \text{Domain}_a \times \text{Domain}_b \rightarrow \text{Order}.
\end{aligned}$$

Definition 30 (Ranking event).

$$\begin{aligned}
q &= (\text{subject}, \text{rater}, \text{behavior}, \\
&\quad \text{evidence}, \text{epoch}, \text{weight}, \sigma). \\
m_q &= \text{subject} \parallel \text{behavior} \parallel \text{evidence} \parallel \text{epoch}, \\
\text{RankValid}(q) &\Leftrightarrow V(k_{\text{rater}}, \sigma, m_q) = \top.
\end{aligned}$$

Definition 31 (Score aggregation).

$$\begin{aligned}
\text{score}_e(s) &= \text{robust}\{\text{weight}(q) \cdot \text{value}(q) : \\
&\quad q.\text{subject} = s\}.
\end{aligned}$$

$$\text{robust} \in \{\text{median}, \text{trimmedMean}, \text{declared}\}.$$

| Attack | Predicate              | Rings control                      |
|--------|------------------------|------------------------------------|
| Sybil  | many DIDs, one actor   | admission cost, ranking, challenge |
| Replay | stale valid $\sigma$   | nonce, TTL, session epoch          |
| MITM   | wrong key for DID      | DID proof, signed envelope, E2E    |
| DoS    | unbounded resource use | quotas, ranking, chunk bounds      |

Table II  
Security predicates and controls

Obligation 2 (Replay rejection).

$$\begin{aligned}
\text{accept}(s, \eta, t) &\Rightarrow \eta \notin \text{Seen}_s \wedge t < \text{exp}(s), \\
\text{Seen}'_s &= \text{Seen}_s \cup \{\eta\}, \\
\text{emitDelivery} &< \text{Seen}'_s.
\end{aligned}$$

### X. Verification and Evaluation

Definition 32 (State relation).

$$\begin{aligned}
C &= (S, Q) \\
Q &= a :: Q_0 \quad I_A(a) = i \quad \tau(S, i) = (S', O, \beta) \\
&\quad \frac{}{(S, Q) \rightarrow (S', Q_0 \cdot \beta)} \\
\text{Explore} &= \text{TLAStyle}(\rightarrow)
\end{aligned}$$

The exploration relation follows TLA+ style and Startertight execution [9], [10].

Invariant 6 (Effect refinement).

$$\begin{aligned}
\text{Impl}_A(s, i) &\preceq I_A^\#(\tau(s, i)). \\
\pi_{\text{state}}(\text{Impl}_A) &= \pi_{\text{state}}(I_A^\# \circ \tau).
\end{aligned}$$

Obligation 3 (Topology coverage).

$$\begin{aligned}
\mathcal{L} &= \{\text{line}, \text{wrap}, \text{gap}, \text{partition}, \text{merge}, \text{churn}\} \\
\mathcal{S} &= \{\text{loss}, \text{dup}, \text{reorder}, \text{timeout}, \text{restart}\}. \\
\text{CheckCase} &= (L, S, P), \quad L \in \mathcal{L}, \quad S \in \mathcal{S}.
\end{aligned}$$

Definition 33 (Latency decomposition).

$$T_{total} = T_{conn} + h \cdot T_{hop} + T_{crypto} + T_{app} + T_{witness}.$$

$$h = O(\log |N|) \quad (F_n \text{ populated}).$$

| Fraction of failed nodes | Mean routing hops | Mean num. of lookup timeouts |
|--------------------------|-------------------|------------------------------|
| 0                        | 3.84              | 0.0                          |
| 0.1                      | 4.03              | 0.60                         |
| 0.2                      | 4.22              | 1.17                         |

Table III  
Latency on failed node rate of Chord [4]

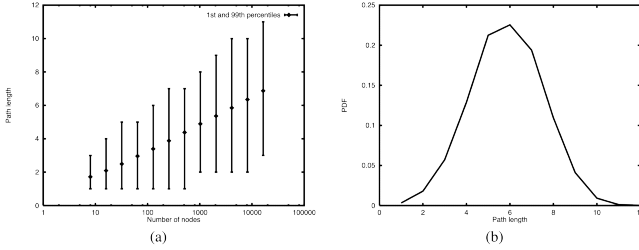


Figure 8. Chord lookup path [4]

Definition 34 (Reliability baseline).

$$failrate = \frac{|F|}{|Q|},$$

$$timeoutMean = \frac{1}{|Q|} \sum_{q \in Q} to(q).$$

| join/leave rate<br>(per sec./stab.<br>period) | Mean<br>routing<br>hops | Mean<br>num. of<br>lookup<br>timeouts | Lookup<br>failures<br>per 10k |
|-----------------------------------------------|-------------------------|---------------------------------------|-------------------------------|
| 0.05/1.5                                      | 3.90                    | 0.5                                   | 0                             |
| 0.10/2                                        | 3.83                    | 0.11                                  | 0                             |
| 0.15/4.5                                      | 3.84                    | 0.16                                  | 2                             |
| 0.20/6                                        | 3.81                    | 0.23                                  | 5                             |
| 0.25/7.5                                      | 3.83                    | 0.30                                  | 6                             |
| 0.30/9                                        | 3.91                    | 0.34                                  | 8                             |
| 0.35/10.5                                     | 3.94                    | 0.42                                  | 16                            |
| 0.40/12                                       | 4.06                    | 0.46                                  | 15                            |

Table IV  
Failure rate of Chord [4]

successor-list extension. The paper therefore uses Chord's circular order as the base model, then uses Zave's CorrectChord results to avoid treating the original Chord maintenance protocol as already correct under churn [4], [5].

Relay anonymity systems take a different route. Tor and Nym route traffic over selected relay paths and focus on hiding communication metadata from local observers or intermediate services [14], [15]. Rings does not replace these systems as anonymity systems. It instead specifies a structured ownership substrate on which relay, mailbox, and hidden-service style protocols can be written. This distinction matters: a relay path can carry traffic, but it does not define the algebraic owner of a DID, mailbox, chunk, or service descriptor.

Blockchain and account-based decentralized systems provide another partial answer. A ledger can publish names, order events, and make writes globally auditable, but the price is global consensus on operations that often only need local routing, replicated storage, or protocol-specific ordering. Rings keeps global consensus outside the base overlay. When a protocol needs stronger ordering evidence, it uses sidecar witnesses rather than changing the overlay into a blockchain.

Browser-native networking is also part of the design space. WebRTC gives browsers a standardized peer transport, while WebAssembly gives portable code a browser execution target [16], [17]. Rings treats these as adapters for the same abstract protocol object, not as separate networks. The same DID/session, Chord owner, effect, and resource-placement laws should hold for browser nodes and native nodes; only the IO interpreter changes.

| System        | Metric   | Ownership law       |
|---------------|----------|---------------------|
| Kademlia fam. | XOR      | bucket routing      |
| Relay mixnets | path     | no structured owner |
| Rings         | circular | owner, Succ, Rep    |

Table V  
Routing and ownership comparison [11]–[15]

Obligation 4 (Evaluation morphism).

$$\Phi_{eval} : (h, timeouts, failures)_{Chord} \mapsto (T_{total}, timeoutMean, failrate)_{Rings},$$

$$\Delta\Phi = T_{conn} + T_{crypto} + T_{witness} + T_{app}.$$

## XI. Related Work

Distributed hash tables give Rings its closest technical ancestry. IPFS and libp2p use Kademlia-derived routing, where distance is measured by XOR and routing state is organized into buckets [11]–[13]. This is a strong fit for content-addressed lookup, but Rings needs a different law: the owner of a DID or VID is the first live node clockwise from that point, and replica ownership is obtained by

## XII. Conclusion

Rings is an algebraic structured peer-to-peer network: structured because ownership, routing, replication, and repair are defined over CorrectChord's circular overlay, and algebraic because each layer preserves an explicit carrier or morphism rather than hiding semantics inside ad hoc network procedures.

$$\mathcal{R} \setminus \setminus f = (R, \mathcal{R}_N, \mathbf{Set}/R, \Omega_N, \mathcal{T}_A^\mathcal{E}, \Pi, \mathbb{G}, \mathcal{X}).$$

|            |   |                                                                                        |
|------------|---|----------------------------------------------------------------------------------------|
| Overlay    | = | $\text{CorrectChord}(R, N, r),$                                                        |
| Resource   | = | $(X_\nu, H_\nu) \in \mathbf{Set}/R,$                                                   |
| Placement  | = | $\Omega_N(X_\nu, H_\nu),$                                                              |
| Transition | = | $\tau_\nu : S_\nu \times I_\nu \rightarrow Y_\nu,$                                     |
| $Y_\nu$    | = | $\mathbf{T}_{A_\nu}^{\mathcal{E}_\nu}(S_\nu \times O_\nu),$                            |
| Crypto     | = | $\mathbb{G} \vee \mathbb{F}_q, \quad \text{Crypto} \cap R = \emptyset,$                |
| Evidence   | = | $\{\text{figures, tables, algorithms}\}$<br>$\cup \{\text{invariants, obligations}\}.$ |

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